ALICE L WALTON SCHOOL OF MEDICINE

Coagulation Assays

Last updated October 27, 2025 · 18 min read · ScholarRx

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Adapted from: Coagulation Assays

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Interpret common coagulation assays.
2. Explain the use of mixing studies.
3. Discuss the use and physiology of the fibrin degradation product and D-dimer assays.

CASE CONNECTION

PA is a 46-year-old female who comes to the emergency department with chest pain. Her vital signs and physical examination are unremarkable. The electrocardiogram is normal. While she is in the radiology department for a chest x-ray, you ask an intern, "What are you thinking?" She tells you she thinks this is musculoskeletal chest pain and that the probability is very low that this is a pulmonary embolism. "But maybe we should get a CT scan anyway. That'll rule it out for sure," she says.

What test will you recommend to rule out a pulmonary embolism for PA? Consider your answer as you read, and we'll revisit PA at the end of the brick.

GO TO CONCLUSION ↓

What Are Coagulation Assays (LO #1)?

Coagulation assays are tests that evaluate the coagulation cascade, helping to determine where there might be a problem in patients with abnormal bleeding.

If a patient presents with unexplained bleeding—or if we just want to assess how well a patient can clot (before performing surgery, for example)—the two main things to evaluate are the platelets (number and function) and the coagulation cascade. Why? Because if you want to make a blood clot, you need to do two things: make a platelet plug, and seal that plug up with fibrin. Platelet plug formation is called primary hemostasis, and fibrin formation, which occurs through the steps of the coagulation cascade, is called secondary hemostasis. Defects i asis can lead to poor clot formation and excessive bleeding.

Types of Tests

The tests most commonly used to assess the coagulation cascade, that is, how well a patient can form fibrin, include the following:

- Prothrombin time (PT) and international normalized ratio (INR)
- Partial thromboplastin time (PTT, activated PTT [aPTT])
- Thrombin time (TT)

We'll also look at mixing studies and how to measure the products left behind after clots dissolve. Platelet assays are covered in another brick.

Technique

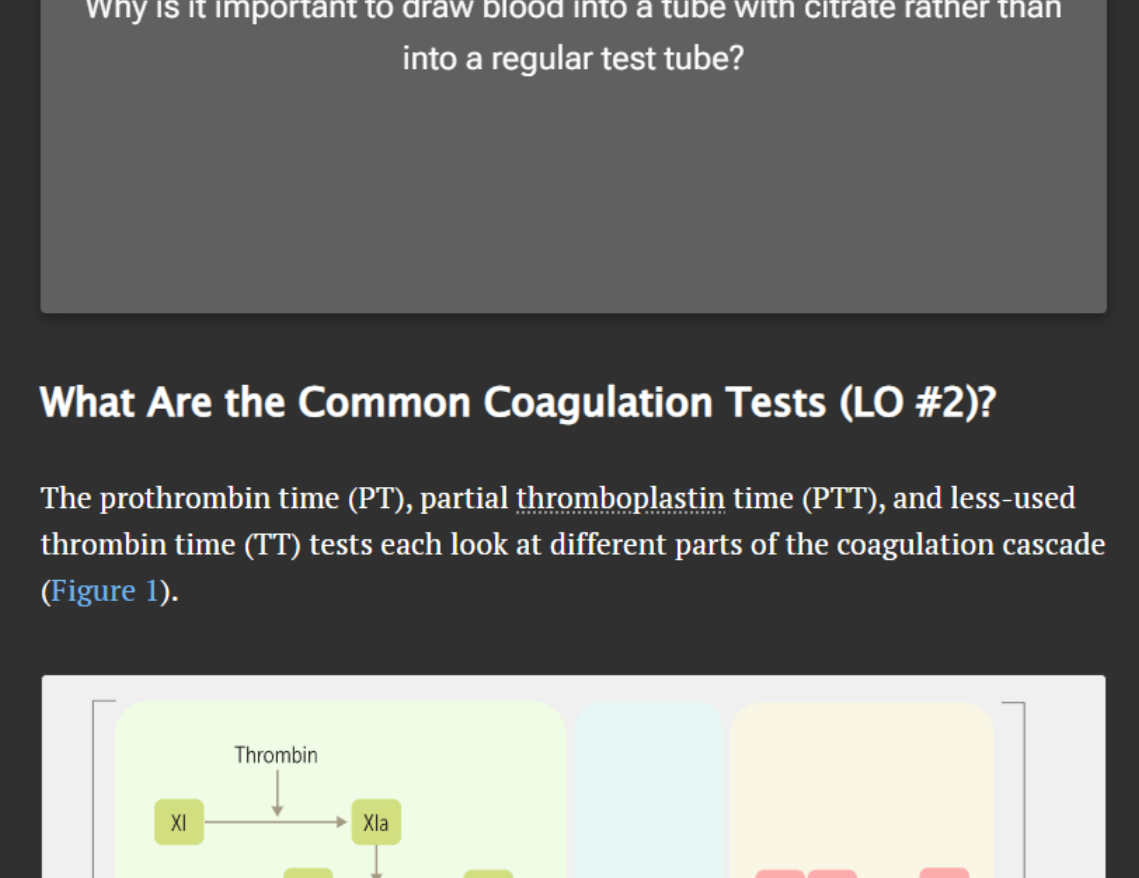
When drawing blood for a coagulation assay, a special vacutainer tube—which usually has a light-blue top—is used. This tube contains citrate, which removes calcium from the blood. Calcium removal prevents clotting, because calcium binds to the phospholipids that appear during platelet activation and provides a surface for assembly of some coagulation factors. This detail is important, because if you just draw blood into a regular test tube and leave it sitting around, it will clot before you get it back to the lab and you will not be able to run the timed tests!

In the lab, the blood is centrifuged, and the plasma layer at the top is collected for testing. At this point, the sample contains just plasma (minus calcium), and it's ready for any coagulation test.

Why is it important to draw blood into a tube with citrate rather than into a regular test tube?

What Are the Common Coagulation Tests (LO #2)?

The prothrombin time (PT), partial thromboplastin time (PTT), and less-used thrombin time (TT) tests each look at different parts of the coagulation cascade (Figure 1).



QUIZ

Figure 1

As a result, they individually give you slightly different pieces of the entire picture, but they complement one another, enabling us to determine how much time it takes your blood to form a clot.

Prothrombin Time and International Normalized Ratio

The prothrombin time (PT) is used to assess the **extrinsic and common pathways** of the coagulation cascade. The PT is determined by taking the patient's plasma, adding thromboplastin reagent (containing phospholipid and a tissue factor–like substance) and calcium, and then measuring the time (in seconds) it takes to form fibrin. As you may recall from your study of normal coagulation, tissue factor combines with factor VIIa to catalyze the conversion of inactive factor X t xtrinsic clotting pathway. So, the PT is a lab measurement of the normal activity of factor VII on down to fibrin formation, on the right side of Figure 1. Note that it will not measure the activities of factors XI, IX, and VIII, which are all part of the intrinsic pathway.

The normal PT range is typically 11 to 15 seconds. The problem with this test is that PT values can fluctuate between hospitals because thromboplastin reagents vary a lot from manufacturer to manufacturer. So, the **international normalized ratio** (INR) was developed to standardize the measurement. The INR is derived from a formula using the ratio of the patient's PT to the PT in serum with normal clotting. As a result, the INR value of the patient will be similar at any hospital. The normal range for the INR is 0.8 to 1.2.

CLINICAL CORRELATION

The INR is often used to measure the degree of anticoagulation of patients on the drug warfarin. Warfarin interferes with the γ-carboxylation of vitamin K–dependent clotting factors, including factors II, VII, IX, and X. Because most of these factors are components of the extrinsic and common pathways (which is what the INR measures), that makes the INR the best test to measure warfarin activity and for clinicians to adjust warfarin levels.

Other clinical uses of PT include hereditary Factor VII deficiency and various acquired deficiencies of coagulation (liver disease, Vitamin K deficiency).

Partial Thromboplastin Time

Contrasting with the ime (PTT) is used to assess the **intrinsic and common pathways** of the coagulation cascade (see Figure 1, left). The PTT was named this way when it was discovered that fibrin could form by using just part of the thromboplastin reagent described above in the PT test. It is now known that the part of the thromboplastin reagent used is the phospholipid portion, and no tissue factor-like substance is present. Therefore, no activation of the extrinsic pathway occurs, as happens for the PT and INR.

The test is done by adding a specific chemical activator of the intrinsic pathway, for example, silica, kaolin, or celite. The PTT will then measure the activity of the intrinsic and common pathways—basically, all the clotting factors except VII, which is part of the extrinsic pathway.

The PTT is determined by taking the patient's plasma, adding phospholipid and calcium, and then measuring the time (in seconds) it takes to form fibrin. This makes sense because normally in the body, tissue factor is necessary to activate the extrinsic pathway, but it is not necessary to activate the intrinsic pathway.

The **activated partial thromboplastin time (aPTT)** is similar to the PTT and is more commonly used in the clinical setting. In the aPTT, an activator is added to the PTT assay, which helps speed up clot formation and results in a narrower reference range. The aPTT is considered to be more sensitive than is the PTT. The normal range of the aPTT is 25 to 40 seconds.

CLINICAL CORRELATION

The aPTT is often used to measure the degree of anticoagulation in patients on the drug heparin. Hepari . Once antithrombin is activa ny factors on the intrinsic and extrinsic pathways, including IIa, VIIa, IXa, Xa, XIa, and XIIa. In practice, the aPTT is much more useful for monitoring the intrinsic and common pathways, so it is used to monitor heparin, but not warfarin, therapy. An anti-Xa assay is now beginning to gain popularity over aPTT for monitoring heparin therapy.

Other clinical uses of aPTT include hereditary factor deficiencies (Factor VIII, IX, XI, and XII) and acquired disorders of coagulation (liver disease, lupus anticoagulant, DIC).

CLINICAL CORRELATION

Although Factor XII deficiency prolongs clotting time in vitro (prolonged aPTT), it does not cause clinical bleeding issues, as it is not involved in physiologic coagulation.

Thrombin Time

The least used of these tests, the thrombin time (TT), assesses the conversion of fibrinogen to fibrin. It is determined by taking the patient's plasma, adding thrombin, and measuring the time (in seconds) it takes to form fibrin. The normal range of the thrombin time is 12 to 14 seconds.

Because this is at the very end of the common pathway, the TT has much more limited use than do the PT and PTT. It's a specialized test most often used to determine whether a patient has a fibrinogen deficiency or fibrinogen functional mutation.

Which pathways of the clotting cascade do the PT and PTT assess?

Clinical Uses

Coagulation assays can be used to monitor anticoagulant therapy (e.g., warfarin and heparin, discussed above). But, another key role is their use for the initial diagnosis of many bleeding disorders involving the coagulation factors. They may be prolonged in genetic disorders, such as hereditary factor deficiencies (e.g., hemophilia), as well as in acquired bleeding disorders resulting from liver disease, vitamin K deficiency, and disseminated intravascular coagulation (DIC).

The most common of these acquired disorders is liver disease. Why? Remember that all of the coagulation factors involved in the coagulation cascade are produced in the liver. Therefore, patients with cirrhosis or other advanced liver diseases can have a h nable to make coagulation factors at a normal rate. In these cases, all coagulation assays are prolonged (PT, PTT, TT) because all the coagulation factors are decreased.

CLINICAL CORRELATION

Another coagulation assay to briefly mention is the **bleeding time.** Bleeding time is when a standardized incision is made on the forearm, and the time it takes for bleeding to stop is measured. Causes of a prolonged bleeding time include: thrombocytopenia (low platelet count), platelet dysfunction, Von Willebrand disease, and vessel wall defects. This test has largely been replaced by other assays, including platelet function tests, as bleeding time has poor accuracy and reproducibility. You may still see it mentioned on national board exams, though!

What Is a Mixing Study (LO #3)?

Causes of Prolonged PTT

In conducting a mixing study, we are trying to determine which of the following is prolonging the PTT:

- A **clotting factor deficiency** (e.g., hemophilia)
- The presence of an **anticoagulant drug** (e.g., heparin)
- The presence of **circulating antibodies that inhibit clotting.** These include autoantibodies such as antiphospholipid antibodies and the lupus anticoagulant. These autoantibodies can occur spontaneously (primary) or in conjunction with autoimmune diseases such as systemic lupus.

Note that the autoan ause abnormal bleeding. Why not? In vitro, these autoantibodies can bind to, and inactivate, reagents that contain phospholipids (such as the thromboplastin reagent that is used in the PTT assay). This makes less reagent available, and in the case of the PTT, it leads to a prolonged PTT. This is a pure lab artifact, and there is no increased bleeding. In fact, in vivo, the presence of autoantibodies like antiphospholipid antibodies or lupus anticoagulant usually causes **increased** thrombosis (clotting), putting patients at risk for deep venous thrombosis and pulmonary embolism!

Conducting and Interpreting the Study

How is the mixing study done? It is performed by mixing the patient's plasma with normal plasma and then running the PTT again.

- **If the new PTT is normal** (if it "corrects"), that means you added something to the mixture (a coagulation factor) that was missing in the patient, so they likely have a clotting factor deficiency (which you can then assay for using special tests).
- **If the new PTT is still prolonged** (if it doesn't correct), that means there is something in the patient's blood (either an anticoagulant drug or an antibody) that is binding up the PTT reagent and falsely prolonging the PTT (Figure 2).
- **If the new PTT corrects into the normal range after mixing but then becomes prolonged,** that means there is a specific factor inhibitor present.

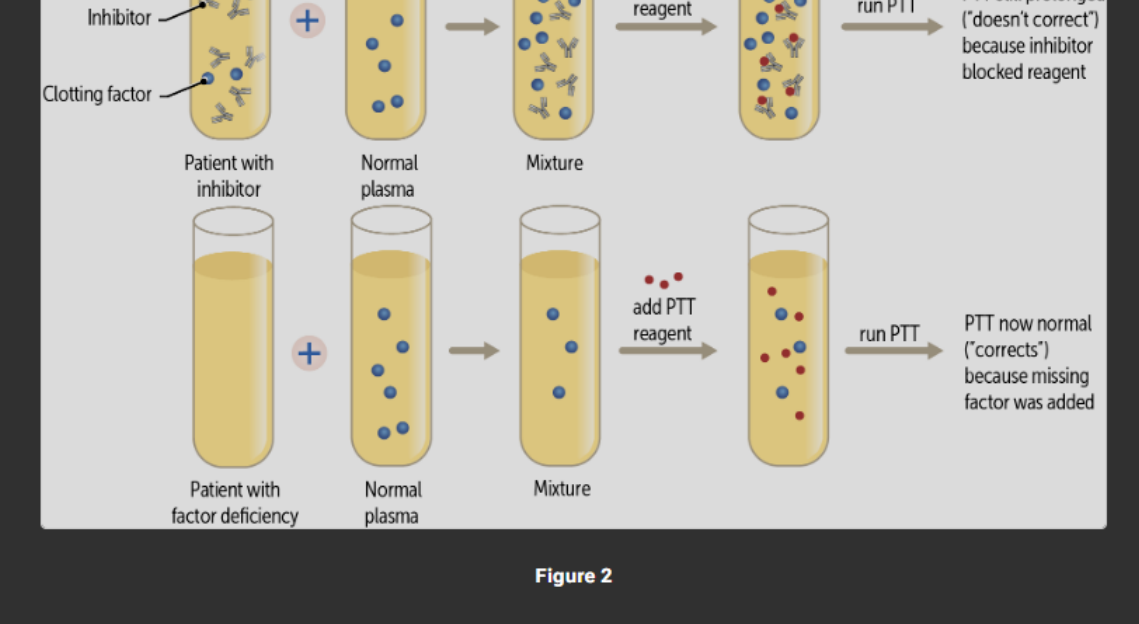

Blood plasma
Inhibitor
Clotting factor
Patient with inhibitor
Normal plasma
Mixture
reagent
run PTT
PTT still prolonged ("doesn't correct") because inhibitor blocked reagent
Patient with factor deficiency
Normal plasma
Mixture
add PTT reagent
run PTT
PTT now normal ("corrects") because missing factor was added

Figure 2

Table 1

	Immediate	1 hr	2 hrs	Interpretations
Case #1	Corrects	Corrects	Corrects	**Factor Deficiency**
Case #2	Corrects	Prolongs	Prolongs	**Factor Inhibitor**
Case #3	Prolongs	Prolongs	Prolongs	**Lupus Anticoagulant**

Why do patients with antiphospholipid antibodies have prolonged PTT?

What Is a Fibrin–Degradation Product Assay (LO #4)?

So far, we have discussed tests for suspected bleeding disorders. What about patients who have excessive clotting (called thrombotic disorders)? Here, there are a couple of useful tests for both diagnosis and monitoring treatment.

The **fibrin degradation product** (**FDP**) **assay** is used to measure the amount of fibrin degradation products present in the serum. These FDPs, also known as split degradation products, are substances released into the serum as a clot dissolves, specifically as fibrin is degraded by plasmin (Figure 3).

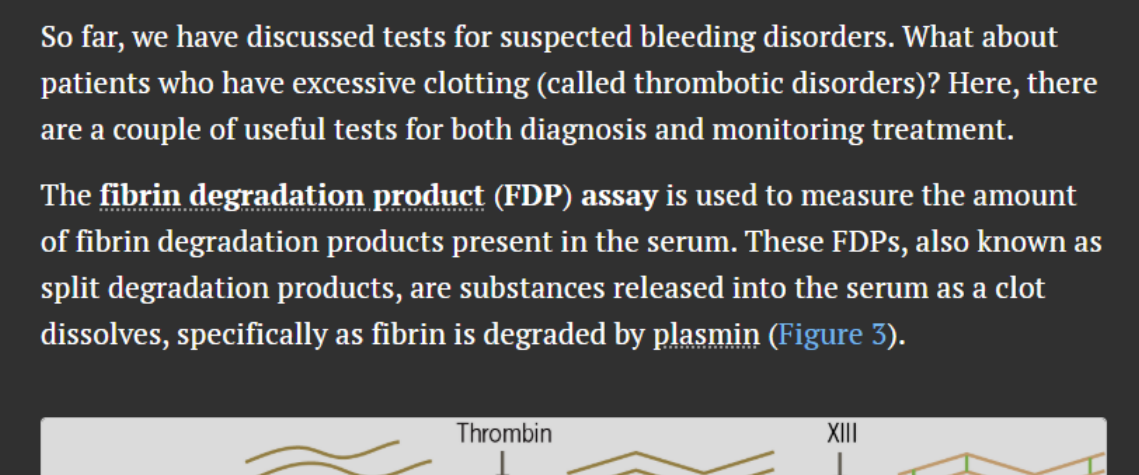


Figure 3

Elevated FDPs indicate that fibrinolysis (breaking down of fibrin, typically in clots) is occurring in the body, implying that there are many clots present (excess thrombosis). The assay has some problems with specificity—note in Figure 3 that FDP can also form from the degradation of fibrinogen (via plasminogen conversion to plasmin catalyzed by tissue plasminogen activator [tPA]), which is the precursor of fibrin and is not a component of clots.

To the rescue comes a cousin of this assay, called the **D-dimer assay**. A D-dimer is a particular type of FDP that forms when cross-linked fibrin is degraded by plasmin (Figure 3, right). So the D-dimer assay measures only chunks of fibrin that have been cross-linked in a clot. This makes it a little more specific for measuring the presence of actual clots, compared with the FDP assay.

Clinical Uses

The FDP and D-dimer assays can be useful in evaluating for disorders with increased clotting and are elevated in deep vein thromboses (DVTs), pulmonary embolism (PE), and disseminated intravascular coagulation (DIC).

CLINICAL CORRELATION

DIC is a disorder se there is abnormal activation of the coagulation cascade, resulting in widespread microthrombi throughout the vessels. This activation leads to the rapid consumption of clotting factors (causing prolonged INR and aPTT), fibrin, and platelets (causing thrombocytopenia). Both hemorrhage and thrombosis occur simultaneously.

Sensitivity

These tests are very sensitive—so sensitive that they can pick up even minute amounts of FDPs or D-dimers. So, they are great at **ruling out** a thrombus because they are almost always elevated if a thrombus is present. Unfortunately, this hypersensitivity means that they could be elevated simply as a result of normal clotting. Because these tests lack specificity, they are not useful for **ruling in** a pathologic thrombus. So, if the tests come back positive, the patient may or may not have a pathologic thrombus and will need more testing. That is usually done with imaging such as ultrasound or angiography.

Why would you be unable to confirm the diagnosis of a pathologic thrombus by using an FDP assay or D-dimer testing?

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to PA, what test do you recommend to rule out a pulmonary embolism?

You now know that the FDP assay and D-dimer test indicate fibrinolysis (breakdown of clots) and chunks of fibrin cross-linked in a clot, respectively. They are highly sensitive tests. You decide to order a D-dimer, and the result is negative. "A negative test result using a highly sensitive test in a low-probability scenario effectively rules out the disease," you think to yourself. You consider this choice part of practicing cost-effective medicine, helping the patient avoid unnecessary testing and saving the system unnecessary expense.

Summary

- Coagulation assays are conducted to determine how well the coagulation cascade works; that is, how well a patient can form fibrin.
- The prothrombin time (PT) and international normalized ratio (INR) measure the activity of the extrinsic and common coagulation pathways.
- The thrombin time (TT) is a specialized test that measures fibrinogen-to-fibrin conversion.
- If the partial thromboplastin time (PTT) is prolonged, a mixing study can help determine whether the prolonged PTT is caused by a factor deficiency or an inhibitor (e ng antibody inhibitor).
- If the rerun PTT is normal, a clotting factor is deficient, but in patients where the PTT remains prolonged, a circulating inhibitor is likely present.
- The fibrin degradation assay measures the products of fibrinolysis (fibrin degradation products).
- The D-dimer assay looks specifically for the degradation product of cross-linked fibrin within a clot and is more specific than is the FDP assay.
- FDP and D-dimer are elevated during excessive clotting but are not specific enough to make the diagnosis of pathologic blood clots; however, a normal test result is useful in ruling out excessive clotting.

Review Questions

1. A patient in the intensive care unit suddenly starts to ooze blood from his central venous line. He is suspected to have disseminated intravascular coagulation. Which of the following lab studies most supports the diagnosis?

- A. ○ High platelet count
- B. ● High partial thromboplastin time that corrects in a mixing study
- C. ○ High partial thromboplastin time that does not correct in a mixing study
- D. ○ Low D-dimer level
- E. ○ Normal prothrombin time

Hide Explanation

The correct answer is high partial thromboplastin time that corrects in a mixing study (B). Disseminated intravascular coagulation (DIC) causes widespread clotting with consumption of clotting factors, prolonging both the prothrombin time (PT) and partial thromboplastin time (PTT), as well as reducing platelets. DIC leads to low platelet counts rather than a high platelet count (A). A mixing study will actually correct this patient's abnormal PT and PTT levels because the mixing study will replenish factor deficiencies associated with DIC (C). DIC also leads to a high plasma D-dimer level, not a low D-dimer level (D). Patients with DIC will have high PTT and high PT, not normal prothrombin time (E), because there is severe thrombocytopenia.

2. A patient with chronic alcohol use disorder arrives to the emergency department intoxicated and presenting with a protuberant abdomen. The patient states that he has been drinking for weeks without a break. Labs show prolonged prothrombin time (PT), partial thromboplastin time (PTT), and thrombocytopenia. Which of the following is the likely cause of the prolonged PT and PTT?

- A. ○ Antibody against factor VII
- B. ○ Chronic kidney disease
- C. ● Cirrhosis
- D. ○ Factor VII
- E. ○ Heparin use

Hide Explanation

The correct answer is cirrhosis (C). This patient likely has cirrhosis, as indicated by their protuberant abdomen and chronic alcohol use. The presence of an antibody against factor VII (A) that would inhibit its function would result in a prolonged PT, not a prolonged PTT. Chronic kidney disease (B) can result in uremia, causing platelet dysfunction, but would not affect clotting factors. Factor VII deficiency (D) would result in a prolonged PT, not a prolonged PTT. Heparin use (E) could elevate both PT and PTT, but the patient is likely not injecting heparin regularly and the clinical picture is more consistent with cirrhosis.

3. A deficiency in which of the following clotting factors would be least likely to cause an elevation in the international normalized ratio?

- A. ○ Factor I
- B. ○ Factor II
- C. ○ Factor VII
- D. ○ Factor X
- E. ● Factor XII

Hide Explanation

The correct answer is factor XII (E). The international normalized ratio (INR) measures activity of the extrinsic and common pathways. Of the listed factors, only factor XII lies in the intrinsic pathway, so it would not be measured in the INR. Factor I (A) and factor II (B) are in the common pathway, so deficiencies would prolong the INR. The same applies to factor VII (C), which is in the extrinsic pathway, and factor X (D), which is in the common pathway.

4. A patient with systemic lupus develops a deep venous thrombosis and is found to have a prolonged partial thromboplastin time (PTT). Which of the following tests is most useful in determining the cause of the prolonged PTT?

- A. ○ Activated partial thromboplastin time
- B. ○ Fibrin degradation product assay
- C. ● Mixing study
- D. ○ Prothrombin time
- E. ○ Thrombin time

Hide Explanation

The correct answer is lupus erythematosus can have autoantibodies (lupus anticoagulant or antiphospholipid antibodies) in their serum, which can in vitro bind to, and inactivate, reagents that contain phospholipids (such as the thromboplastin reagent used in the partial thromboplastin time [PTT] assay). This makes the reagent less available and leads to prolonged PTT (a pure lab artifact). They cause clotting in vivo, however, leading to this patient's deep vein thrombosis. A mixing study is performed by mixing the patient's plasma with normal plasma and rerunning the PTT. If this is normal, a clotting factor is deficient, but in patients with inhibiting autoantibodies, the PTT remains prolonged. The activated partial thromboplastin time (A) is a test similar to the PTT and would not be helpful in showing the presence of an inhibiting antibody. A fibrin degradation product assay (B) would be used as indirect evidence to check whether any clot is present in the body, but it would not demonstrate the autoantibody. Prothrombin time (D) would be used to check for extrinsic and common pathway functioning and may be altered in patients taking warfarin or with liver disease, but it is not useful here. Thrombin time (E) would be used to assess conversion of fibrinogen to fibrin, which may be prolonged in patients with fibrinogen deficiency or a functional mutation.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Interpret common coagulation assays.~~

02 ☑ ~~Explain the use of mixing studies.~~

03 ☑ ~~Discuss the use and physiology of the fibrin degradation product and D-dimer assays.~~

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Objective Confidence: 3/3